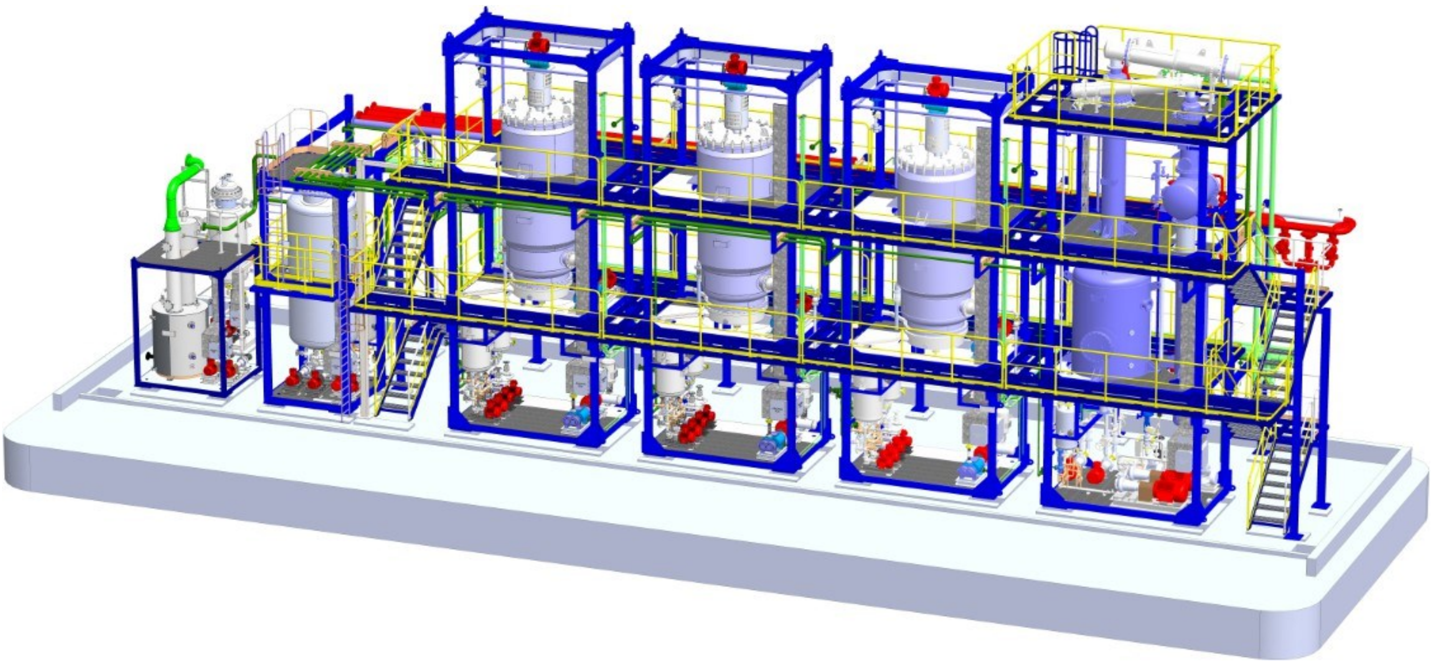
Sub.: Thermopac Technology For Used Oil Re-Refining Project Budgetary Offer For
Engine Oil Refinery PLC SCADA Controlled Automatic Continuous

We are pleased to submit our offer for design, manufacturing, supply and commissioning of used oil re-refining plant to be installed and commissioned.

This technology is developed to process used engine oil to its maximum output by means of environmentally friendly fully automatic continuous process. Our technology uses specially designed **TWFE (Thermopac Wiped Film Evaporator) process to** optimize the yield.

The process and its advantages along with all technical details are described in the following pages.



For **THERMOPAC**
(Turnkey Engineering Solution Division)

Marketing Manager

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Raw Material Required: -

The oil refining plant uses used lubricating oil as a raw material. The used oil can typically come from garages, service stations, mechanic shops, and industrial users. The used oil typically contains water, diluents, solid particles from rust and corrosion, dirt and other suspended and dissolved impurities.

Typical Analysis of Used Lube Oil:

Appearance	:	Viscous liquid with suspended impurities
Color	:	Black
Specific gravity (D-1298)	:	0.850 to 0.900
Water content (% in emulsion w/w)(D-4006)	:	5 to 7%
Flash Point, °C (D-92)	:	100 to 190
Viscosity, cSt at 40°C	:	70 to 110
At 100 °C (D-445)	:	9.5 TO 12.5
Ash sulphated, % w/w (D-482)	:	1.5 to 3.0
Pentane insoluble, % w/w (D-893)	:	1.0
Total acid no. mg KOH/g	:	1.0

The Design Specifications:

The plant is designed based on typical feed characteristics as above and for input specification given below.

Input*:* used oil of mixed hydrocarbons containing:

~ 5 % water
~ 5 - 8% light ends [gasoline, aromatics]
~ 5 - 8% diesel, kerosene
~ 70 - 80% long-chain hydrocarbons
~ 5 - 10% residual solids and asphaltenes.

Output specifications:

From Skid -1:

~ 5% water containing traces [< 2%] of hydrocarbons;
~ 5% - 10 % lights [naphthalene, gasoline, kerosene, and diesel]

From Skid - 2:

~ 50% - 55% medium chain hydrocarbons (neutral 150)
~ 50% - 45% heavy used oil [containing heavy base oil, solids and asphaltenes]

OR

~ 30% - 35% longer chain hydrocarbons (neutral 300)
~ 10% - 15% asphalt extender [containing solids and asphaltenes]

Process:

Our Plant processes used lube oil, using high vacuum molecular distillation technology in the following six stages:

Stage 1: De-hydration and De-gas oil Skid I
Stage 2: Distillation Skid-II
Stage 3: Pollution Skid -IV Scrubber and Thermal Oxidizer
Stage 4: Finishing for Group-I

This process technology re-refines used lube oil to its maximum output by means of environmentally friendly PLC controlled fully automatic, continuous process.
The plant processes used lube oil and industrial oils, using high vacuum molecular distillation technology.
This process is based on three different modular skids for three important stages of process:

Item class A: Standard scope supply

Excluding site fabrication, piping, electrical and civil jobs

Stage 1: Skid - I: De-hydration and De-gas Oil Skid. (Front End Skid)

Removes traces of water, naphtha, diesel oil and gas oil by flash evaporation; and separates water: and diesel, naphtha and gas oil as different distillation cuts and sends all uncondensed gases to pollution skid .

Stage 2: Skid-II) Base Oil Extraction by TWFE (Back End Skid) Dehydrated and free from water, naphthalene and gas oil, the used lube oil will be heated to a specific temperature and sent to a specially designed evaporator to extract light lube oil. This cut is named as processed light lube oil and sent to storage tank for subsequent polishing system to generate Light Base Oil.

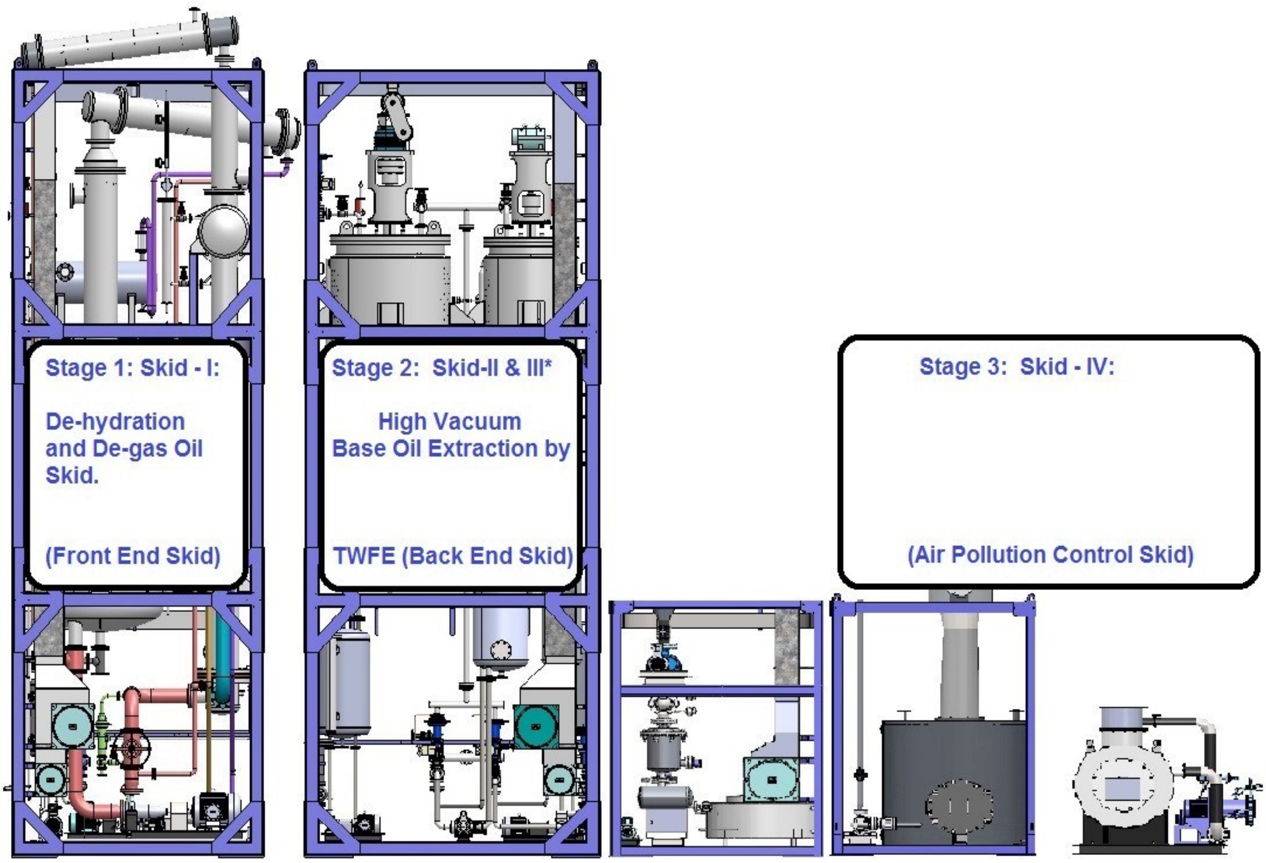
Now the remaining used oil containing heavy lube oil and residue will be sent to specially designed Thermo Wiped Film Evaporator, where the heavier hydrocarbon fractions will evaporate and condense and collect into a receiver. This cut is named as processed heavy lube oil and sent to storage tank for further polishing system to generate Heavy Base Oil.

Bottoms of this Thermo Wiped Film Evaporator is basically residue which will be transferred to asphalt tank for further processing into RFO.

Vacuum system on this skid can maintain desired vacuum and temperatures ; all exhaust from this vacuum system is sent to the pollution skid

Stage 3: Skid - IV: (Pollution Skid)

All vacuum exhaust and dehydration skid vent gases will be scrubbed to absorb SOx and H2S compounds through specially designed scrubbing system and balance exhaust gas will now contain non condensable VOC. This will be sent to thermal oxidation.



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Stage 4: **Fully Automatic Non Consumable Regenerateable Finishing System**

Adds value to reclaimed oils by improving color and reducing or removing odors
The clay polishing systems provides an economical alternative to hydrogen treatment plants as a means of adding value to re-refined lube oil.

The system is based upon banks of active columns that contain adsorptive media. The physical characteristics of the media allow it to be re-activated once saturated, and thereby permitting several hundred cycles to be run before being replaced.

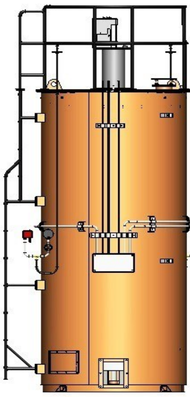
Once exhausted, typically of 6 months to a year of continuous operation, the media is disposed off in a conventional landfill site, as a dry non-hazardous waste.

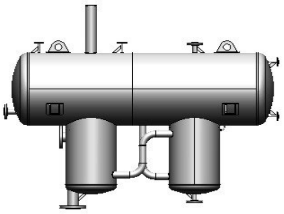
Processing oil through the system results in oil that is stable against oxidation over any measured timescale - this increase in shelf life is an added benefit of the polishing system. Furthermore, the Viscosity Index is not degraded by the polishing system.

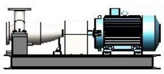
High Temperature Thermal Oil System

The Thermal Oil Heating system consists of the following equipment:

1. High Temp Heater
2. High Temp Circulation Pump
3. Expansion cum Deaerator tank



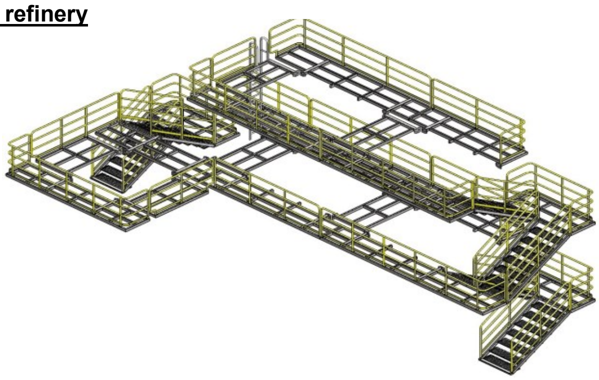




Supply of Thermal Oil fluid is excluded from our standard supply.

Platform, Walkway, Ladder and Equipment Lifting arrangment around refinery

Fixed platforms, walkways, stairways and ladders, complete with:



Guard railing
Fall protection from ladders
Ladder cages and enclosures
Access hatches

Supply of chain hoist is excluded from our standard supply.

Parallel or Series operations can be achieved through the system as it is designed to switch from parallel to series mode operation via the custom-built SCADA supplied with every system. This allows the plant owner to easily switch between different processing modes for each oil stream reclaimed through the front-end system. Integration with existing front-end systems and/or tank farms is easily achieved.

The out product characteristics will completely depend on the feed properties.
The viscosities of the **Light Base Oil** and the **Heavy Base Oil** can be adjusted to desired values by controlling the temperatures and vacuums.
The viscosities of the **Light Base Oil** and the **Heavy Base Oil** fully depend on input feed oil (used oil feed)

We can produce Heavy Oil as heavy as 80 -120 cSt at 40 deg C with flash points ranging from 210 - 240 deg C. However, the output percentage fully depends on long-chain hydrocarbon content of the feed. System is designed to handle 15% free and emulsified water in lube oil along with light ends such as diesel, naphtha and gas oil.

Control and Automation System

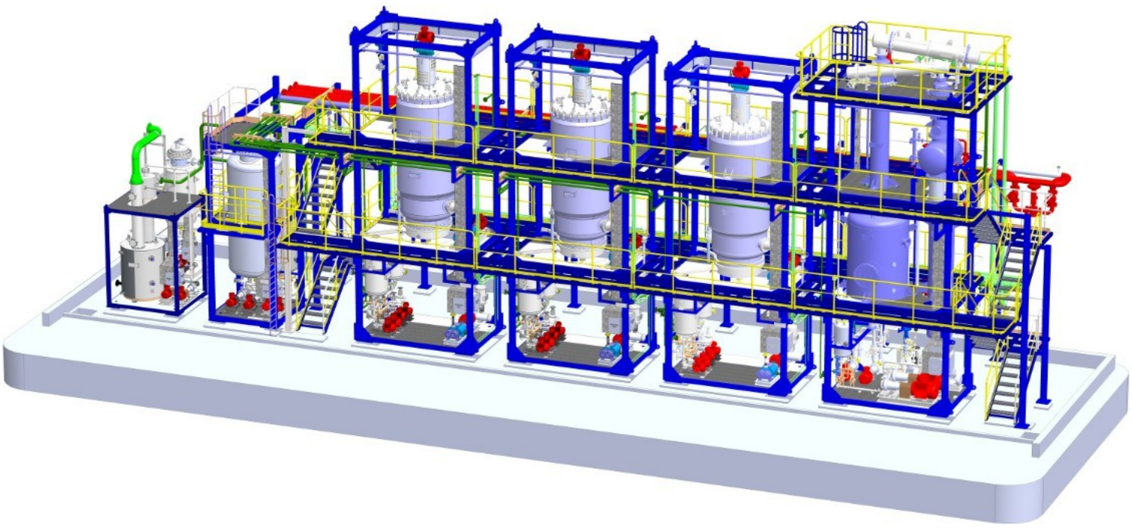
The operation of the Plant is fully automatic and remote controlled by a PLC and Scada, which centralizes all control and monitoring systems to allow Plant operation in a steady, reliable and safe manner.

Air compressors, Thermal Oil Heater, Thermal Oxidizer and Cooling tower will be controlled by PLC and Scada.

Controller configuration memory will have battery backup so that the controller maintains its configuration, state and information in the event of power failure.

OUR SKID MOUNTED UNITS COMPRISE:

All skids are pre-wired and are complete with necessary flame proof instruments and controllers such as temperature, pressure, flow and level and local junction boxes. All hot surfaces are insulated and clad wherever required. Input and output piping along with control and one isolation valve wherever required is part of our supply.



External refinery piping shown in above model is excluded from our standard supply,

OUTPUT PRODUCT SPECIFICATIONS:

Typical specification of the output products

- Light Lube SN 150

	Refined base oil prior to finishing
Viscosity @ 40 deg cSt.	~25 - 37
Flash Point deg C – closed cup	>180°C
Density @ 15 deg C, kg/l	~ 0.85

- Heavy Lube SN 400-500

	Refined base oil prior to finishing
Viscosity @ 40 deg cSt	~80 -110
Flash Point – closed cup	>230°C
Density @ 15 deg C	~ 0.88

BY-PRODUCTS

The Plant will also produce the by-products with the typical characteristics as follows. The actual characteristics will be depending on the feedstock.

1. Light Fuel

Specific gravity	0.760 - 0.770
Flash point, °C	20

2. **Gasoil**

Specific gravity	0.840 - 0.860
Viscosity, cSt at 40°C	3 - 6
Flash point, °C	55 min
TAN, mg KOH/gr	0.1
LHV, kcal/kg	10,000

3. **Asphaltic Residue**

Specific gravity	0.900 - 1.050
Viscosity, cSt @100°C	300
Ash content, wt%	5 - 10
Sulfur, wt%	1 - 2
Softening point, °C	10 - 15
LHV, kcal/kg	8,500
Metals, wt%	2 - 4

OR

4. 180 CST Fuel Oil

Furnace oil is a dark viscous residual product used as a fuel
In different types of combustion equipment.
kinematic viscosity range of 125 to 180 CSt at 50°C

OUR SERVICES

- Documentation
  - Plot Plans and General Arrangement and PFD's
  - Material and Heat Balances
  - Equipment Data Sheets and Specification Sheets
  - P&I's and Piping Specs and Layout drawings
  - Package Units Specifications
  - Instrumentation Drawings
  - Electrical Single Line Diagrams, Electrical Drawings
  - Insulation & Painting Specifications
  - Operating Manuals
  - Mechanical Catalogues.
- Our Exact Scope in Construction, Erection, Pre-commissioning and Commissioning.
  - Process design and technical know-how
  - Mechanical design, engineering and structural basic design.
  - Electrical basic design
  - Instruments and control systems basic design.
  - Procurement and supply of the plants and equipments
  - Supervision of construction, erection/installation of the plant
  - Startup / commissioning of the plant
  - Operation / training to the staff.
- Pre-commissioning and Commissioning Assistance:
  - 4-week onsite assistance for final hookup of utilities and instruments and control wiring.
  - 2-week onsite assistance for startup
  - 2-week onsite assistance for demonstration and training.

MAJOR EMISSIONS & EFFLUENTS

- a) Emission from thermic fluid heater chimney.
- b) Very light ends storage tank and Skid vents pass through thermal oxidation system
- c) Distilled water recovered from the used oil containing traces (<2%) of hydrocarbons should be routed to a used water storage and/or treatment system (Out of our scope)

KEY EQUIPMENTS OF THE PROCESS

Equipment Description	Country of Origin
Evaporator	Thermopac India
Wiped Film Evaporator	Thermopac India
Vacuum Pumps	Korea
Vacuum Booster	USA
Pumps	India
High Temperature Pumps	KSB India
Thermic Fluid Heater	Thermopac India
Mechanical Seals	Eagle, India
Switchgear	India
Control Panel	Thermopac India

PROCESS GUARANTEES:

- Guaranteed throughput in the range of +/- 10%
- Asphalt viscosity @ 30°C, min > 1000 cSt
- Guaranteed 75% yield of base oil
- Maximum 15% yield of asphalt
- Fuel oil consumption, maximum 0.05 kg/kg ULO

CODES AND STANDARDS

- American Society of Testing Materials (ASTM)
- American Society of Mechanical Engineers (ASME)
- American Petroleum Institute (API)
- Tubular Exchanger Manufacturer's Association (TEMA)
- Local Standards where the plants will be located

ADVANTAGES OF TWFE PROCESS

- Sludge free & 100% environment friendly
- Highly efficient latest technology
- PLC automated, with SCADA system
- Skid mounted - Expandable, reduces set-up cost and time
- Modular construction
- Handles a variety of used lube oils (ULO)

SHIPMENT DETAILS

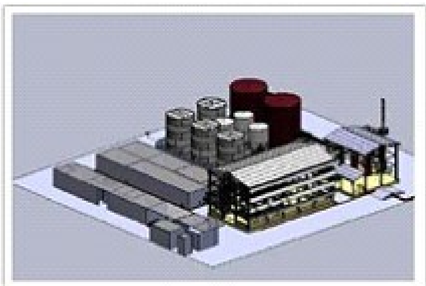
We will be shipping the entire project from our factory near Mumbai, India. Even when the price quoted is on Ex-works basis (nearest port to your plant location), further transportation to your project site, loading, unloading, leading and placing of the equipment on foundations are to be arranged by the client at his cost.

INFRASTRUCTURE

Given below is tentative refinery land and connected electrical load.

Capacity and Infrastucture

Model	Connected Load (KW)	Water Requirement LPD/GPD	Minimum land Size Sq.Mts / Sq. Ft.
UOR-750	200	3000/790	6000/65000
UOR-1000	200	4000/1000	6000/65000
UOR-1500	325	6000/1500	10000/110000
UOR-2000	400	9000/2400	10000/110000
UOR-3000	575	12000/3100	15000/160000
UOR-4500	850	18000/4750	15000/160000
UOR-7500	1,100	30000/8000	25000/265000
UOR-10000	1,250	40000/10500	40000/430000



SCHEDULE AND UNSCHEDULED MAINTENACE

The plant should be taken up for cleaning / greasing/ oiling on schedule maintenance as follows:

Schedule	No. of days	Total days in year
Monthly	1	12
Quarterly	4	16
Unscheduled		10
Total down time days in a year		38 Days

SPARE PARTS:

Essential spare parts required for 1 year of trouble free operation of refinery and equipment's is part of our standard supply.

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TIME SCHEDULE OF THE PROJECT

- 5 to 6 Months for shipment
- 1.5 Months shipping time
- 1 Month for Erection, piping, testing and commissioning.
- 1 Month contingency allowance

Total time Total time from issuing the P.O with advance to commissioning of plant *9.5 **months**

*The schedule as above is subject to force majeure as well as order backlog at the time of finalization of order.

WARRANTY:

All plant and equipment are guaranteed for performance as per specification provided in the offer and we will demonstrate the performance of the plant for 48 hours during the trial run.

Performance demonstration will include:

- Throughput capacity
- Product and byproduct specification
- Consumption of utilities

All performance figures on capacity, product specifications, and utility consumption are subject to a tolerance of ±10%

All components of the plant are covered under limited warranties given by our suppliers or by us

All components and plant are guaranteed by THERMOPAC for period of 12 Months from date of commissioning or 18 months from date of shipment whichever is earlier.

Above warranty covers replacement or repair of components at free of cost and does not cover any liability towards any damage to personnel, property, or loss in production.

Process Flow Diagram: